

FENT2321

Wind Energy and Design of a Wind Turbine

Assignment 3: Preliminary Blade Design and Design-Point Rotor Performance

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1 Design parameters

Parameter	Value	Unit
Number of blades, Z	2	–
Design wind speed, V_1	12.0	m/s
Design tip-speed ratio, λ_{des}	4.0	–
Rotor radius, R	0.30	m
Number of blade elements, N	15	–
Initial chord at $r/R \approx 0.7$	50	mm
Air density, ρ	1.2	kg/m ³
Dynamic viscosity, μ	1.47×10^{-5}	kg/(m s)
Reynolds convergence tolerance	30000	–
Airfoil	NACA 2414	–

Table 1: Design parameters used for the preliminary blade design.

The rotational speed corresponding to the design point is

$$\omega = \frac{\lambda_{\text{des}} V_1}{R} = \frac{4.0 \cdot 12}{0.30} = 160 \text{ rad/s.} \quad (1)$$

2 Airfoil selection and aerodynamic data

I chose to continue with the NACA 2414. NACA 2414 has a maximum thickness of approximately 14%, which is consistent with the initial thickness guideline.

The available aerodynamic data used in the calculation are the values of C_L , C_D , and α_{opt} corresponding to the maximum C_L/C_D at each Reynolds number.

Re	C_L	C_D	α_{opt} [deg]
100000	0.9155	0.01903	6.50
200000	0.8850	0.01424	6.00
500000	0.8155	0.01027	5.00
1000000	0.8797	0.00977	5.75

Table 2: NACA 2414 aerodynamic data used in the BEM calculation.

For Reynolds numbers between the available polar data points, linear interpolation is used to get accurate values used in the BEM calculations.

3 BEM method

3.1 Blade element discretization

The blade is divided into

$$N = 15$$

elements of width

$$dr = \frac{R}{N} = \frac{0.30}{15} = 0.020 \text{ m.} \quad (2)$$

The local radius is taken at the centre of each element,

$$r_i = \left(i - \frac{1}{2}\right) dr, \quad i = 1, \dots, N. \quad (3)$$

For $N = 15$, one element lies exactly at

$$r/R = 0.700.$$

This element is therefore used for the Step 1 and Step 2 Reynolds-number calculation.

3.2 Step 1: Initial Reynolds number

Starting with the assumed values

$$a = \frac{1}{3}, \quad a' = 0, \quad c = 0.050 \text{ m}.$$

At $r/R = 0.7$, the relative velocity is

$$W = \sqrt{[(1-a)V_1]^2 + [(1+a')\omega r]^2}. \quad (4)$$

The resulting value is approximately

$$W = 34.54 \text{ m/s}.$$

The first Reynolds-number estimate is therefore

$$Re_{\text{first}} = \frac{\rho W c}{\mu} \approx 1.410 \times 10^5. \quad (5)$$

The value used in the calculation is

$$\boxed{Re_{\text{first}} = 140977}.$$

Since this value lies between $Re = 100000$ and $Re = 200000$, the corresponding aerodynamic parameters are obtained by linear interpolation. The resulting values are approximately

$$C_L = 0.9030, \quad C_D = 0.01707, \quad \alpha_{\text{opt}} = 6.295^\circ.$$

3.3 Step 2: Reynolds-number iteration

At $r/R = 0.7$, the local tip-speed ratio is

$$\lambda_r = \frac{r}{R} \lambda_{\text{des}} = 0.7 \cdot 4.0 = 2.8. \quad (6)$$

The axial induction factor is obtained by solving

$$16a^3 - 24a^2 + a(9 - 3\lambda_r^2) - 1 + \lambda_r^2 = 0. \quad (7)$$

The tangential induction factor is

$$a' = \frac{1 - 3a}{4a - 1}, \quad (8)$$

and the flow angle is

$$\phi = \tan^{-1} \left(\frac{1 - a}{(1 + a')\lambda_r} \right). \quad (9)$$

The axial force coefficient is

$$C_a = C_L \cos \phi + C_D \sin \phi. \quad (10)$$

The chord is then calculated from

$$\frac{L_c}{R} = \frac{8\pi a(r/R) \sin^2 \phi}{(1-a)ZC_a}. \quad (11)$$

The relative velocity and new Reynolds number are

$$W = \sqrt{[(1-a)V_1]^2 + [(1+a')\omega r]^2}, \quad (12)$$

$$Re_{\text{new}} = \frac{\rho W L_c}{\mu}. \quad (13)$$

The iteration is stopped when

$$|Re_{\text{new}} - Re_{\text{old}}| \leq 30000. \quad (14)$$

The iteration history is shown in Table 3.

Table 3: Reynolds-number iteration at $r/R = 0.7$.

Iteration	Re_{old}	C_L	C_D	α_{opt} [deg]	Re_{new}
1	140977	0.90300	0.01707	6.295	219181
2	219181	0.88056	0.01399	5.936	224925

The second iteration satisfies the convergence criterion because

$$|224925 - 219181| = 5744 < 30000.$$

The converged Reynolds number is therefore taken as

$$\boxed{Re = 224925}.$$

Using the converged Reynolds number to interpolate the airfoil data gives

$$C_L = 0.87923, \quad (15)$$

$$C_D = 0.01391, \quad (16)$$

$$\alpha_{\text{des}} = 5.917^\circ, \quad (17)$$

$$\frac{C_L}{C_D} = 63.21. \quad (18)$$

4 Preliminary blade design

Once the Reynolds-number iteration has converged, the resulting C_L , C_D , and α_{des} are used for the preliminary blade design. The local values of λ_r , a , a' , ϕ , and C_a are calculated separately for every blade element.

The twist angle is calculated as

$$\theta = \phi - \alpha_{\text{des}}. \quad (19)$$

The resulting blade geometry and induction factors are shown in Table 4.

Table 4: Preliminary blade geometry, induction factors, and local Reynolds number.

r/R	L_c/R	θ [deg]	a	a'	Re
0.033	0.1983	49.020	0.2673	2.8567	52153
0.100	0.4201	39.549	0.2919	0.7417	122642
0.167	0.4873	31.623	0.3066	0.3535	162972
0.233	0.4812	25.400	0.3154	0.2056	186275
0.300	0.4482	20.620	0.3207	0.1335	200251
0.367	0.4088	16.941	0.3241	0.0932	209038
0.433	0.3709	14.071	0.3264	0.0685	214823
0.500	0.3370	11.793	0.3279	0.0524	218795
0.567	0.3073	9.954	0.3290	0.0413	221624
0.633	0.2817	8.444	0.3298	0.0333	223701
0.700	0.2595	7.186	0.3304	0.0275	225269
0.767	0.2402	6.123	0.3309	0.0230	226479
0.833	0.2234	5.216	0.3312	0.0196	227433
0.900	0.2087	4.432	0.3315	0.0168	228197
0.967	0.1956	3.750	0.3317	0.0146	228818

The corresponding physical chord lengths range from approximately 58.7 mm at the outermost element to a maximum of approximately 146.2 mm around $r/R = 0.167$. To me, this sounds large, but I struggle to fully understand what I am doing here so it maybe wrong.

4.1 Chord distribution

Figure 1 shows the resulting non-dimensional chord distribution.

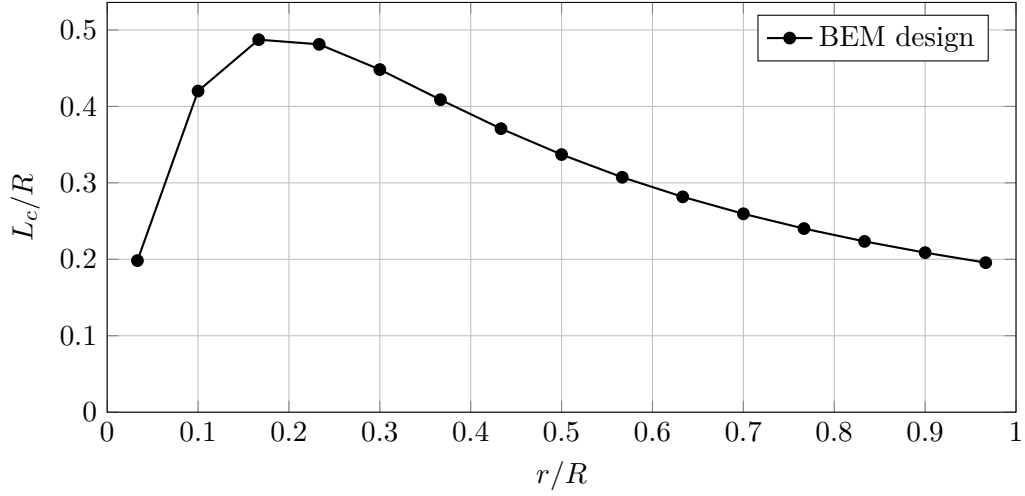


Figure 1: Calculated chord distribution along the blade.

4.2 Twist distribution

Figure 2 shows the resulting twist distribution.

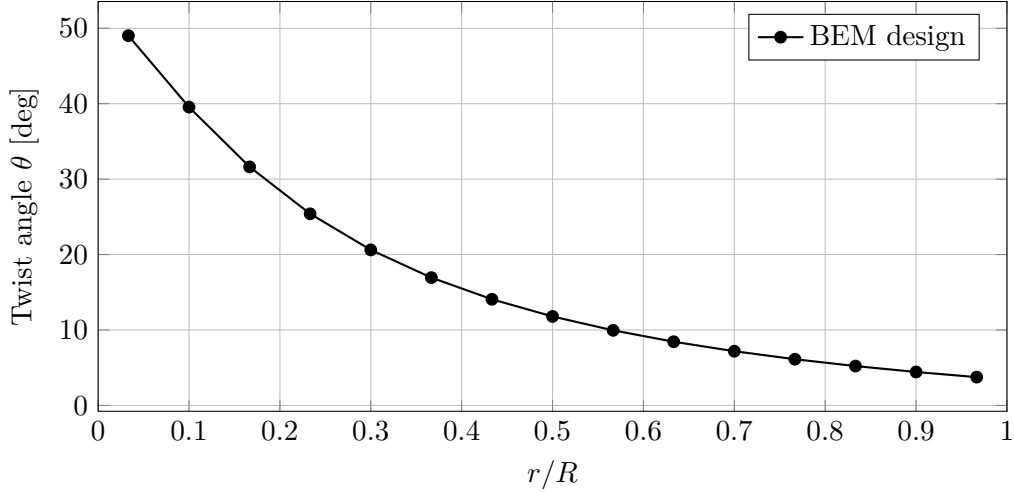


Figure 2: Calculated twist distribution along the blade.

5 Discussion of blade design

5.1 Induction factors

The ideal axial induction factor is $a = 1/3$ and the ideal tangential induction factor is $a' = 0$. The calculated values approach these ideal values as the radius increases.

In the outer part of the blade, approximately from $r/R = 0.3$ to the tip, a is between about 0.321 and 0.332. This is close to the ideal value of $1/3$. The axial induction factor is lower near the root, reaching approximately $a = 0.267$ at the innermost element.

The tangential induction factor shows a much stronger deviation near the root. It decreases from $a' = 2.86$ at $r/R = 0.033$ to approximately $a' = 0.0146$ at $r/R = 0.967$. It decreases close to zero as we move outwards on the wing.

The large root-region deviations are therefore consistent with the assignment's warning that the ideal induction factors are mainly expected over most of the blade rather than at the root.

5.2 Reynolds-number variation along the blade

The Reynolds number varies substantially over the blade. The calculated local value increases from approximately

$$Re = 5.2 \times 10^4$$

at the innermost element to

$$Re = 2.29 \times 10^5$$

at the outermost element.

At $r/R = 0.7$, the local value is approximately

$$Re = 2.25 \times 10^5,$$

which is very close to the converged Step 2 value of

$$Re = 2.249 \times 10^5.$$

This agreement is expected because Step 2 was specifically performed at $r/R = 0.7$. The Reynolds number is lower than the Step 2 value in the inboard region and slightly higher toward the tip.

An important limitation is that the innermost calculated Reynolds numbers fall below the lowest available airfoil polar Reynolds number of 10^5 . However, Step 3 is prescribed to use the converged aerodynamic parameters from Step 2 along the blade, so these local Reynolds numbers are reported as the actual resulting values rather than being used to update the aerodynamic coefficients.

5.3 Effect of assuming constant Reynolds number

The Step 3 design uses the converged aerodynamic parameters from $r/R \approx 0.7$ as representative values along the entire blade. In reality, the Reynolds number changes from approximately 5.2×10^4 near the root to 2.3×10^5 near the tip.

If the aerodynamic coefficients were instead updated locally using the actual Reynolds number, the chord and twist distributions could change, particularly in the inboard region where the difference from the representative Reynolds number is largest. The effect is expected to be most significant near the root because the local Reynolds number is well below the Reynolds number used for the Step 2 design. The outer part of the blade is less affected because its Reynolds number is close to the converged value.

This is therefore a simplification of the preliminary design procedure. A locally Reynolds-number-dependent BEM calculation could provide a more realistic blade design, but this is listed as an optional extension in the assignment.

5.4 Comparison with a commercial wind-turbine blade

The calculated chord distribution has a pronounced inboard region with a large chord, followed by a gradual reduction in chord toward the tip. This general tapering behaviour is qualitatively similar to a practical wind-turbine blade, where the chord normally becomes smaller toward the tip.

However, the present distribution should not be interpreted as a manufacturable blade geometry. The chord increases strongly from the first element to a maximum around $r/R = 0.17$, and the root region is governed by the idealized BEM equations. A commercial blade would normally have a structurally designed root, a smoother transition from the root section, and additional practical constraints. The assignment explicitly treats the hub as visualization only at this stage, and production constraints are handled later in Assignment 4.

Thus, the present chord distribution captures the aerodynamic tapering trend but should be regarded as a preliminary aerodynamic design rather than a final commercial blade geometry.

6 Forces acting on the blade

For each blade element, the normal and tangential force coefficients are calculated from

$$C_n = C_L \cos \phi + C_D \sin \phi, \quad (20)$$

$$C_t = C_L \sin \phi - C_D \cos \phi. \quad (21)$$

The element thrust and tangential force for all Z blades are calculated using

$$Z dT' = Z \left(\frac{1}{2} \rho W^2 c C_n dr \right), \quad (22)$$

$$Z dM' = Z \left(\frac{1}{2} \rho W^2 c C_t dr \right). \quad (23)$$

The torque contribution from each element is

$$dQ = r(Z dM'). \quad (24)$$

The calculated force distribution is given in Table 5.

Table 5: Thrust and tangential force acting on all two blades for each blade element.

r/R	$Z dT' \text{ [N]}$	$Z dM' \text{ [N]}$
0.033	0.085	0.117
0.100	0.269	0.265
0.167	0.462	0.343
0.233	0.656	0.385
0.300	0.852	0.409
0.367	1.047	0.422
0.433	1.241	0.429
0.500	1.436	0.434
0.567	1.630	0.436
0.633	1.824	0.436
0.700	2.018	0.436
0.767	2.211	0.435
0.833	2.405	0.434
0.900	2.599	0.432
0.967	2.792	0.430

The thrust force increases toward the blade tip because the local relative velocity increases strongly with radius. The tangential force reaches a fairly broad maximum in the mid-to-outer part of the blade and then changes only slightly toward the tip.

7 Design-point rotor performance

The total rotor thrust and torque are obtained by summing the blade-element contributions:

$$T = \sum_i Z dT'_i, \quad (25)$$

$$Q = \sum_i r_i(Z dM'_i). \quad (26)$$

The total power is

$$P = \omega Q. \quad (27)$$

The power and thrust coefficients are calculated from

$$C_P = \frac{P}{\frac{1}{2}\rho AV_1^3}, \quad (28)$$

$$C_T = \frac{T}{\frac{1}{2}\rho AV_1^2}, \quad (29)$$

where

$$A = \pi R^2.$$

The resulting design-point performance is

Table 6: Design-point rotor performance.

Quantity	Value	Unit
Total thrust, T	21.526	N
Total torque, Q	0.9559	N m
Power, P	152.94	W
Power coefficient, C_P	0.5217	–
Thrust coefficient, C_T	0.8812	–